## 0.1 Ideal

**Definition 0.1.1.** Let $R$ be a Ring. A subset $I \subseteq R$ is called *ideal* of $R$ if:

1. $I \subseteq R$ is a subgroup of $R$.
2. $I$ is closed under the multiplication.
3. For any $r \in R$, $rI \subseteq I$ and $Ir \subseteq I$. (In other word, for any $r \in R, a \in I$, $ra \in I$ and $ar \in I$.)

**Theorem 0.1.2.** Let $R$ be a Ring. Then, TFAE:

1. $I \subset R$ is an Ideal of $R$.
2. The additive Quotient Group $R/I \overset{\text{def}}{=} \{r + I \mid r \in R\}$ be a Ring under the operation:

$$(r + I) \times (s + I) = (rs) + I$$

*Proof.* Observation:

$$r_1 + I = r_2 + I \iff r_1 - r_2 \in I \iff \exists a \in I \ \text{ s.t } \ r_1 = r_2 + a$$

Now, for well-definedness, want to show that the equality

$$\begin{aligned} &(r + I) \times (s + I) = (rs) + I \\ \overset{(*)}{=} &[(r + \alpha) + I] \times [(s + \beta) + I] = (r + \alpha)(s + \beta) + I = (rs + r\beta + \alpha s + \alpha\beta) + I \end{aligned}$$

$(*)$ holds for any $r, s \in R$, $\alpha, \beta \in I$.
If $I$ is Ideal, then $r\beta, \alpha s, \alpha\beta \in I$. Thus closed under the addition gives $(*)$.
Conversely, if this operation is well-defined, then for any $r, s \in R$, $\alpha, \beta \in I$, $(*)$ holds.
Substituting zero to each $r, s, \alpha, \beta$ gives $I$ is ideal. □

### 0.1.1 Properties of Ideal in Ring with identity

**Definition 0.1.3.** Let $R$ be a Ring with identity, and $A \subseteq R$. Define *Ideal generated by* $A$ as:

$$(A) \overset{\text{def}}{=} \bigcap_{\substack{I \text{ ideal} \\ A \subseteq I}} I$$

And,

$$RA \overset{\text{def}}{=} \{r_1 a_1 + \cdots + r_n a_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\}$$
$$AR \overset{\text{def}}{=} \{a_1 r_1 + \cdots + a_n r_n \mid n \in \mathbb{N}, r_i \in R, a_i \in A\}$$
$$RAR \overset{\text{def}}{=} \{r_1 a_1 r_1' + \cdots + r_n a_n r_n' \mid n \in \mathbb{N}, r_i, r_i' \in R, a_i \in A\}$$

**Lemma 0.1.4.** Let $R$ be a Ring with identity, and $A \subseteq R$. Then, $(A) = RAR$.

*Proof.* Since $RAR$ is ideal which contains $A$, $(A) \subseteq RAR$.
And, conversely, if $\sum_{i=1}^{n} r_i a_i r_i' \in RAR$, then $\sum_{i=1}^{n} r_i a_i r_i' \in (A)$ because each $r_i a_i r_i'$ are contained in $(A)$,
being $(A)$ is ideal containing $A$ and ideal is closed under the addition. □

**Theorem 0.1.5.** Let $I$ be an ideal of Ring $R$ with identity.

$$I = R \text{ if and only if } I \text{ contains a unit.}$$

*Proof.* Right direction is clear by $1 \in R = I$.
Denote $u \in I$ be a unit with $vu = 1$, and Let $r \in R$ be given. Then,

$$r = r1 = rvu \in I$$

□

**Definition 0.1.6.** An Ideal $M$ of $R$ is *Maximal ideal* if: There is no Ideal $I$ such that $M \subsetneq I \subsetneq R$.

**Theorem 0.1.7.** Let $R$ be a Ring with identity.
Then, every proper ideal $I \subsetneq R$ is contained in a maximal ideal.

*Proof.* □

Prop. Let $I, J \subset R$ be ideals of a Ring $R$ with unicity.
1) $I+J$ is the smallest ideal of $R$ containing both $I$ and $J$.
2) $IJ$ is an ideal contained in $I \cap J$.
3) If $R$ is Commutative and $I+J=R$, then $IJ=I \cap J$.

Proof. The first and second, ask a toddler on the street.

Let $a \in I \cap J$. Since $1 \in R = I+J$, $1 = b+c$ for some $b \in I$, $c \in J$, so

$$a = a \cdot 1 = a(b+c) = ab + ac = \underset{\in IJ}{ba} + \underset{\in IJ}{ac}.$$

Thus $I \cap J \subseteq IJ$.

**35.** Let $I$, $J$, $K$ be ideals of $R$. (with 1)

**(a)** Prove that $I(J+K) = IJ + IK$ and $(I+J)K = IK + JK$.

**(b)** Prove that if $J \subseteq I$ then $I \cap (J+K) = J + (I \cap K)$.

**Lemma 0.1.8.** Let $R$ be a commutative Ring with identity, $M, P$ are proper ideals of $R$.

1. $M$ is Maximal Ideal *if and only if* $R/M$ is a field.
2. $P$ is Prime Ideal *if and only if* $R/M$ is an integral domain.

Summary: $M$ maximal $\iff$ $R/M$ field $\implies$ $R/M$ integral domain $\iff$ $M$ prime.

*Proof.*

$$
\begin{aligned}
M \text{ is maximal} &\iff \text{There is no ideal } I \text{ such that } M \subsetneq I \subsetneq R \\
&\iff \text{There are Ideals of } R/M \text{ only } 0 \text{ and } R/M \\
&\iff R/M \text{ is field}
\end{aligned}
$$

$$
\begin{aligned}
P \text{ is Prime Ideal} &\iff \text{If } ab \in P, \text{ then } a \in P \text{ or } b \in P \\
&\iff \text{If } ab + P = P, \text{ then } a + P = P \text{ or } b + P = P \\
&\iff \text{If } \overline{a}\overline{b} = \overline{0}, \text{ then } \overline{a} = \overline{0} \text{ or } \overline{b} = \overline{0} \\
&\iff R/P \text{ is integral domain}
\end{aligned}
$$

□